

1. 椭球 $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

$x = a \cos \theta \sin \varphi, y = b \sin \theta \sin \varphi, z = c \cos \varphi$

θ 表示绕 z 轴旋转的角度

φ 表示从北极到南极的纬度角

对于固定的 $\varphi = \varphi_0$, 我们可以得到一个平面上的曲线

对于不同的 φ , 我们可以得到一系列等高截面

当 $z = c \cos \varphi_0$.

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \sin^2 \varphi_0$$

线性变换 $x' = \frac{x}{a}, y' = \frac{y}{b}$.

$$x'^2 + y'^2 = \sin^2 \varphi_0$$

覆盖性: $\varphi \in [0, \pi]$ 表示从北极到南极的所有纬度. 在每个固定的 $\varphi = \varphi_0$ 平面上都存在一个圆 (通过线性变换得到) $\therefore$ 覆盖了所有点

圆周的不相交性: 不同的 $\varphi = \varphi_0$ 对应不同的截面高度, 每个高度上的圆周互不相交, 且彼此独立

2. 单叶双曲面.

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$

固定 $z = z_0$.

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 + \frac{z_0^2}{c^2}$$

$$x' = \frac{x}{a \sqrt{1 + \frac{z_0^2}{c^2}}}, y' = \frac{y}{b \sqrt{1 + \frac{z_0^2}{c^2}}}$$

$$x'^2 + y'^2 = 1$$

双叶双曲面:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = -1$$

固定 $z = z_0$.

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z_0^2}{c^2} - 1$$

$$x' = \frac{x}{a \sqrt{\frac{z_0^2}{c^2} - 1}}, y' = \frac{y}{b \sqrt{\frac{z_0^2}{c^2} - 1}}$$

$$x'^2 + y'^2 = 1$$

二次锥面:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 0$$

固定 $z = z_0, \frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z_0^2}{c^2}$

$$x' = \frac{x}{a \sqrt{\frac{z_0^2}{c^2}}}, y' = \frac{y}{b \sqrt{\frac{z_0^2}{c^2}}}$$

$$x'^2 + y'^2 = 1$$

3. 椭圆抛物面

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z}{c}$$

固定 $z = z_0$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z_0}{c}$$

$$x' = \frac{x}{a\sqrt{\frac{z_0}{c}}} \quad y' = \frac{y}{b\sqrt{\frac{z_0}{c}}}$$

$$x'^2 + y'^2 = 1$$

椭圆柱面

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$x' = \frac{x}{a} \quad y' = \frac{y}{b}$$

$$x'^2 + y'^2 = 1$$

$$4. 2x^2 + 3y^2 + 4z^2 - xy - yz = 1$$

$$\text{二次项 } 2x^2 + 3y^2 + 4z^2 - xy - yz$$

$$[x \ y \ z] \begin{bmatrix} 2 & -\frac{1}{2} & 0 \\ -\frac{1}{2} & 3 & -\frac{1}{2} \\ 0 & -\frac{1}{2} & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad \text{特征向量} \quad \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} \quad \begin{bmatrix} 2\sqrt{6}+5 \\ \sqrt{6}+2 \\ 1 \end{bmatrix} \quad \begin{bmatrix} -2\sqrt{6}+5 \\ -\sqrt{6}+2 \\ 1 \end{bmatrix} \quad P = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{2\sqrt{6}+5}{6+2\sqrt{6}} & \frac{-2\sqrt{6}+5}{6-2\sqrt{6}} \\ \frac{2}{\sqrt{6}} & \frac{\sqrt{6}+2}{6+2\sqrt{6}} & \frac{-\sqrt{6}+2}{6-2\sqrt{6}} \\ \frac{1}{\sqrt{6}} & \frac{1}{6+2\sqrt{6}} & \frac{1}{6-2\sqrt{6}} \end{bmatrix}$$

$$A = P \Lambda P^T$$

$$[x \ y \ z] P \Lambda P^T \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = 1$$

$$\text{令 } [x' \ y' \ z'] = [x \ y \ z] P$$

$$\text{则 } [x' \ y' \ z'] \Lambda \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = 3x'^2 + \frac{6-\sqrt{6}}{2}y'^2 + \frac{6+\sqrt{6}}{2}z'^2 = 1$$

$$[x \ y \ z] = [x' \ y' \ z'] \begin{bmatrix} \frac{-1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{2\sqrt{6}+5}{6+2\sqrt{6}} & \frac{\sqrt{6}+2}{6+2\sqrt{6}} & \frac{1}{6+2\sqrt{6}} \\ \frac{-2\sqrt{6}+5}{6-2\sqrt{6}} & \frac{-\sqrt{6}+2}{6-2\sqrt{6}} & \frac{1}{6-2\sqrt{6}} \end{bmatrix} \quad \dots \quad \left[\frac{1}{6-2\sqrt{6}} \right] x' + \frac{1}{6-2\sqrt{6}} z'$$

$$f = x + y + z = \frac{2}{\sqrt{6}}x' + \frac{2\sqrt{6}+8}{6+2\sqrt{6}}y' + \frac{8-2\sqrt{6}}{6-2\sqrt{6}}z'$$

$$f^2 = \left(\frac{\sqrt{6}}{3}x' + \frac{3\sqrt{6}+8}{6+2\sqrt{6}}y' + \frac{8-2\sqrt{6}}{6-2\sqrt{6}}z' \right)^2 = \left[\left(\frac{\sqrt{6}}{3}x' \right)^2 + \left(\frac{6-\sqrt{6}}{2}y' \right)^2 + \left(\frac{6+\sqrt{6}}{2}z' \right)^2 \right] \left[\left(\frac{\sqrt{6}}{3} \right)^2 + \left(\frac{3\sqrt{6}+8}{6+2\sqrt{6}} \right)^2 + \left(\frac{8-2\sqrt{6}}{6-2\sqrt{6}} \right)^2 \right]$$

$$= 1 \cdot \frac{64}{15}$$

$$f \in \left[-\frac{8}{15}\sqrt{15}, \frac{8}{15}\sqrt{15} \right]$$

$$5. \text{ 与 } 2x^2 + 3y^2 + 4z^2 - xy - yz = 1 \text{ 相切的直线为}$$

$$x + y + z + \frac{8}{15}\sqrt{15}$$

$$\therefore \text{最小距离} = \frac{100 - \frac{8}{15}\sqrt{15}}{\sqrt{3}}$$